

Questions on Operators in Python

1. Create variables for your name, age, and city, and print a formatted sentence using those variables.
2. Store an integer, float, string, and boolean in separate variables and print their data types.
3. Assign the value 10 to variable a, then increment it by 5 using the += operator and print the result.
4. Swap two variables x = 5 and y = 10 using a temporary variable, and print the swapped values.
5. Swap two variables without using a third variable and print the result.
6. Multiply a string "hello" by 3 using the * operator and print the result.
7. Create a bill calculator that multiplies price and quantity and prints the total cost.
8. Calculate the perimeter and area of a rectangle using variables length and width.
9. Convert Celsius to Fahrenheit using the formula and arithmetic operators only.
10. Calculate the average of three numbers and print it using arithmetic operators.
11. Use the // operator to show floor division between two integers (e.g., 100 // 3).
12. Use the % operator to find the remainder when a number is divided by another (e.g., 25 % 4).
13. Show the difference between regular division / and floor division // using two numbers.
14. Store a float number and convert it to an integer using typecasting. Print both.
15. Store a string "123" and convert it to an integer. Multiply by 2 and print the result.
16. Use the ** operator to raise 2 to the power of 5 and print the result.
17. Store a list of fruits and use the in operator to check if "apple" is in the list (no if-statement).
18. Use the is operator to check if two variables point to the same memory location.
19. Store two booleans: True and False. Print the result of their and, or, and not operations.
20. Use the not in operator to check if "grape" is **not** present in a list of fruits.

21. Create three variables: `a = 2`, `b = 3`, and `c = 4`. Use one expression to compute `a * b + c` and print the result.
22. Store your first and last name in two variables. Concatenate them with a space in between.
23. Demonstrate the use of `*=` by multiplying a number by 10 and printing the new value.
24. Create a variable `score = 10`, add 5 to it, then subtract 2 from it using `+=` and `-=` operators. Print the final score.
25. Create a bill with three items and add up their prices using only `+=` on a total variable.
26. Show how left shift (`<<`) and right shift (`>>`) operators work on the number 4.
27. Use the `^` operator to perform bitwise XOR on two integers and print the result.
28. Use a tuple unpacking to assign three values to three variables and print them.
29. Create a string and use the `len()` function to count its length, store it in a variable, and print it.
30. Store `a=10`, `b=10.0` and compare them using `==` and `is`, then print both results.